THIAGO SAUD

Senior AI & MLOps Engineer — Principal Front-End Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer (7+ years) and Specialist in Architecture of high-performance and critical front-end systems, with a proven history of direct financial impact on ecosystems that move trillions of reais and have millions of active customers (BTG Pactual & XP Investimentos). Reports directly to C-Level and acts with technical mentoring of engineers. In active transition to AI Engineer & MLOps, combining technical rigor in the development of Machine Learning, Deep Learning, GenAI, LangChain, LangGraph pipelines and RAG architectures.

CORE COMPETENCIES

- **AI, MLOps, GenAI, LLMs & Computer Vision:** *Machine Learning · Deep Learning · Computer Vision · Facial Detection and Recognition · AR/VR · PyTorch · TensorFlow · Keras · Scikit-learn · Hugging Face · MLflow · AWS SageMaker · Azure ML · IBM Watsonx · LangGraph · LangChain · LlamaIndex · RAG · Fine-tuning (LoRA/QLoRA) · ChromaDB · FAISS · ONNX · NLU/NLP · MCP · Autonomous Agents · Prompt & Harness Engineering.*
- **Architecture & Design Patterns:** *Domain-Driven Design (DDD) · Test-Driven Development (TDD) · Clean Architecture · SOLID · Design System · System Design · Micro-frontends · Clean Code.*
- **Data Engineering & Backend: FastAPI:** *FastAPI · SQL · NoSQL · PostgreSQL · REST · GraphQL · WebSockets · UV · TOML.*
- **DevOps, SecOps & Quality: Docker:** CI/CD · *Docker · AWS (S3, CloudFront, Cloud9, CloudShell) · GitHub Actions · Azure DevOps · SonarQube · Semgrep · Gitleaks · Bandit · Git Hooks · Datadog · Sentry · Core Web Vitals (LCP, INP & CLS) · Playwright · Vitest · Jest · Karma · Pytest · Semver · Linters · Prettier · NGINX · Apache · Jenkins · OAuth 2.0/JWT · OWASP Top 10.*
- **Core Stack:** *Python · TypeScript · JavaScript (ES6+) · Java, C/C++ · Angular · React · Vue.js · Next.js · Vite · Stencil.js · Storybook · Redux Saga· RxJS · NgRx · React Query · Vuex · Vuetify · Angular Material.*
- **Technical Leadership & Processes:** *Mentoring · Code Review · ADR & RFC · Brag Document · Agile Methodologies (Scrum/Kanban).*

PROFESSIONAL EXPERIENCE

BANCO BTG PACTUAL S.A. — **São Paulo, SP, BR** ***(REMOTE)***

Principal Front-End Engineer · Digital Equities Vertical | **Feb 2023 – Jun 2026** ***(3 years and 5 months)***

- Hired directly in *5 months (previously via outsourced), 2 years earlier than expected,* due to *strategic impact, technical leadership and ownership.*
- Led the architecture of *white-label financial platforms (Home Broker Web & COE)* and *mentored 5 software engineers*, for high-frequency operations at *B3*, contributing part of *4.8 million* customers, *R$ 1.6 trillion* in *AUM*, with revenue growth of *37% YoY* and *ROAE* of *28.1% (3Q2025).*
- Designed *component governance (Nx/Turborepo, Nexus, Storybook & Angular)* and evolved the corporate *Design System (ORQUESTRA)*, increasing code reuse by *60%* and reducing visual defects by *35% (WCAG/A11Y).*
- Spearheaded advanced rendering *(GPU, Virtualization & Paging)* and reactivity optimizations *(Angular Signals & RxJS)*, cutting page load times by *45%* and ensuring *60 FPS* fluidity under real-time high-volatility data.
- Adopted *AI Engineering* across the development, *DevOps, and SecOps* lifecycle, reducing delivery time by *65%* and code defects by *80%* via *LLMs, MCP, Cursor/Copilot, Autonomous Agents & Prompt & Harness Engineering.*
- Deployed advanced *micro-frontends architecture (Webpack Module Federation)* backed by top-tier observability and security *(Datadog, Sentry & OWASP Top 10).*

XP INVESTIMENTOS S.A. — **São Paulo, SP, BR** ***(REMOTE)***

Senior Front-End Engineer · Global Markets Vertical · Treasury | **Jul 2021 – Nov 2022** ***(1 year and 5 months)***

- Architected critical corporate financial platforms *(Black Sales & Black AAI)*, supporting *12,000* advisors and *4.5 million* active customers, *R$ 1.2 trillion* in assets and *R$ 19.8 billion* in consolidated gross revenue *(4Q2023).*
- Defended complex architectural guidelines *(RFCs & ADRs)* directly to the *Engineering Manager and C-Level (CFO/Treasury)*, standardizing *100%* of legacy transition workflows and accelerating technical approvals by *40%.*

- Implemented a *scalable micro-frontends architecture (Webpack Module Federation),* high-performance reactive flows *(React, GraphQL, Redux-Saga, DDD & Clean Architecture)* and advanced rendering optimizations (GPU & Virtualization & Pagination).
- Evolved the *corporate Design System (SOMA)*, boosting component reuse by *50%* and mitigating visual bugs by *33% (WCAG/A11Y)*.

ZOOX TECNOLOGIA LTDA. — **Rio de Janeiro, RJ, BR** ***(REMOTE)***
Mid-Level Front-End Engineer | **Jul 2020 – Nov 2020** ***(4 months)***

- Acted as a technical reference in *hospitality PMS white-label platform* with direct *stakeholder* management and strategic alignment.
- Modeled critical *facial recognition flows (Face matching & Bounding boxes)* via *AWS Rekognition*, achieving *>90%* accuracy and *100%* regulatory compliance with *LGPD* and *FNRH (Ministry of Tourism)*.
- Structured real-time billing architecture integrating *global OTAs (Booking.com, Expedia & Agoda)* via *PostgreSQL*, eliminating *overbooking* operational failures and executing automated *check-in/out* transactions for thousands of daily rooms with *Vuex*.

TG4 TECNOLOGIA LTDA. — **Rio de Janeiro, RJ, BR** ***(ON-SITE)***
Junior Front-End Engineer | **Sep 2019 – Jul 2020** ***(10 months)***

- Reported directly to *CTIO* in the development of the *white-label chatbot* for *5 telecom operators (Tim, Claro, Vivo, Oi & Nextel)*.
- Integrated *NLU models* and *predictive recommendation algorithms (Next Best Action)*, doubling upgrade offer acceptance rates by *75%* and driving key KPIs *(CAC, CPC & LTV)* via *(DDD, Clean Architecture, Angular, Docker & Playwright).*

NEORIS DO BRASIL LTDA. — **Rio de Janeiro, RJ, BR** ***(HYBRID)***
R&D Technology Intern | **Apr 2017 – Apr 2019** ***(2 years)***

- Acted as a sole intern reporting directly to *3 Senior Managers* across *2 strategic large-scale products* for the *client Aliansce (Shopping Center Administrator)* and 2 *complex internal PoCs.*
- Developed the *"Aliansce BI Geospatial" web platform (Vanilla Javascript & Google Maps API)* and *2 native Android AR/VR apps (Java & Vuforia SDK)*, alongside *computer vision* prototypes *(Microsoft Azure Cognitive Services)*.

AI & MACHINE LEARNING PERSONAL PROJECTS

- **CAUSAL TRANSFORMER MODEL (SHAKESPEARE):** *Pipeline* in *TensorFlow and Keras* for stylized text generation (Shakespearean literary tone) with *custom tokenization*, *checkpointing (TensorBoard)*, *sampling (Temperature/Top-k)* and *Gradio*. Built with *dependency injection*, *strict typing* and 100% *test coverage via pytest.*
- **TECHNICAL REGRESSION PIPELINE (BYD):** Advanced statistical predictive modeling *(linear, multiple, and logistic regression)* with *scikit-learn* and *pandas* applied to automotive specs. Automated generation of visual analytical reports and validation metrics *(R^2 & Precision)* tested with *pytest*.
- **AUTONOMOUS AGENTS ECOSYSTEM:** Proprietary multi-layer *Python agent architecture* featuring *Meteorological Agent (OpenWeather API with persistent memory), Restaurant Agent (PDFs & TF-IDF cosine similarity)* and *Intelligent Router.* 100% test coverage via *pytest.*

EDUCATION & CERTIFICATIONS OF PROFESSIONAL SPECIALIZATIONS

- **BACHELOR OF COMPUTER SCIENCE** — **Rio de Janeiro, BR** ***(ON-SITE)***
  Universidade Veiga de Almeida (UVA) | **Jan 2014 – Feb 2019** ***(5 years and 2 months)***
  Capstone: *"Facial Recognition for Public Safety" (Python, OpenCV, Viola-Jones & PCA/LDA; Accuracy ~80–100%, response time ~0.8s).*

- **BUSINESS ADMINISTRATION AND MANAGEMENT** — **Spain, Catalonia** ***(REMOTE)***
  Universitat de Barcelona (UB) | **Jan 2018 – Nov 2019** ***(1 year and 11 months)***

- **CERTIFICATIONS:** ***GOOGLE: Foundations of User Experience Design (2024)*** · ***UNIVERSITY OF MICHIGAN: Python (2026)*** · ***ANTHROPIC****: AI Fluency: Framework & Foundations (2026)* · ***DEEPLEARNING.AI: Deep Learning (2026)*** · ***IBM: AI Engineering (2026)*** · ***DUKE UNIVERSITY****: MLOps | Machine Learning Operations (2026).*

- **LANGUAGES:** Portuguese *(Native)* · English & Spanish *(Limited working* proficiency*).*